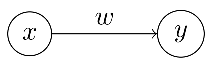
1 Twice example

1.1 Model



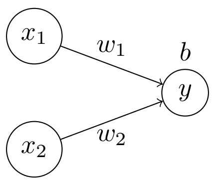
$$y = x \cdot w$$

1.2 Cost

$$\begin{aligned} C &= (xw - y)^2 \\ \frac{dC}{dw} &= \frac{d}{dw}(xw - y)^2 \\ &= 2(xw - y)x \end{aligned}$$

2 Gates example

2.1 Model



$$\begin{aligned} y &= \sigma(x_1 w_1 + x_2 w_2 + b) \\ \sigma(x) &= \frac{1}{1 + e^{-x}} \\ \frac{d}{dx}\sigma(x) &= \sigma(x)(1 - \sigma(x)) \end{aligned}$$

2.2 Cost

$$a = \sigma(x_1 w_1 + x_2 w_2 + b)$$

$$\begin{aligned} \frac{\partial a}{\partial w_1} &= \frac{\partial}{\partial w_1}\sigma(x_1 w_1 + x_2 w_2 + b) \\ &= \sigma(x_1 w_1 + x_2 w_2 + b)(1 - \sigma(x_1 w_1 + x_2 w_2 + b))\frac{\partial}{\partial w_1}(x_1 w_1 + x_2 w_2 + b) \\ &= a(1 - a)x_1 \end{aligned}$$

$$
\begin{aligned}
\frac{\partial a}{\partial w_2} &= \frac{\partial}{\partial w_2}\sigma(x_1w_1 + x_2w_2 + b) \\
&= \sigma(x_1w_1 + x_2w_2 + b)(1 - \sigma(x_1w_1 + x_2w_2 + b))\frac{\partial}{\partial w_2}(x_1w_1 + x_2w_2 + b) \\
&= a(1-a)x_2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a}{\partial b} &= \frac{\partial}{\partial b}\sigma(x_1w_1 + x_2w_2 + b) \\
&= \sigma(x_1w_1 + x_2w_2 + b)(1 - \sigma(x_1w_1 + x_2w_2 + b))\frac{\partial}{\partial b}(x_1w_1 + x_2w_2 + b) \\
&= a(1-a)
\end{aligned}
$$

$$
C = (a_i - y_i)^2
$$

$$
\begin{aligned}
\frac{\partial C}{\partial w_1} &= \frac{\partial}{\partial w_1}(a-y)^2 \\
&= 2(a-y)\frac{\partial}{\partial w_1}(a-y) \\
&= 2(a-y)a(1-a)x_1
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial w_2} &= \frac{\partial}{\partial w_2}(a-y)^2 \\
&= 2(a-y)\frac{\partial}{\partial w_2}(a-y) \\
&= 2(a-y)a(1-a)x_2
\end{aligned}
$$

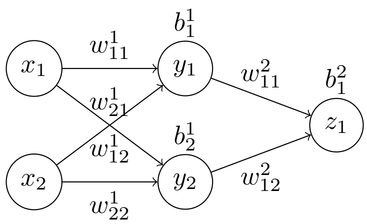
$$
\begin{aligned}
\frac{\partial C}{\partial b} &= \frac{\partial}{\partial b}(a-y)^2 \\
&= 2(a-y)\frac{\partial}{\partial b}(a-y) \\
&= 2(a-y)a(1-a)
\end{aligned}
$$

2.3 Cost - Simplified

$$
\begin{aligned}
a &= \sigma(x_1 w_1 + x_2 w_2 + b) \\
\frac{\partial a}{\partial w_i} &= a(1-a)x_i \\
\frac{\partial a}{\partial b} &= a(1-a) \\
C &= (a-y)^2 \\
\frac{\partial C}{\partial w_i} &= 2(a-y)a(1-a)x_i \\
\frac{\partial C}{\partial b} &= 2(a-y)a(1-a)
\end{aligned}
$$

3 Xor example

3.1 Model



$$
\begin{aligned}
&y_1 = \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
&y_2 = \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&z_1 = \sigma(y_1 w_{11}^2 + y_2 w_{12}^2 + b_1^2) \\
&\sigma(x) = \frac{1}{1+e^{-x}} \\
&\frac{d}{dx}\sigma(x) = \sigma(x)(1-\sigma(x))
\end{aligned}
$$

3.2 Cost

$$
\begin{aligned}
a_1^1 &= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
a_2^1 &= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
a_1^2 &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^1}{\partial w_{11}^1} &= \frac{\partial}{\partial w_{11}^1}\sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
&= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1)(1-\sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1))\frac{\partial}{\partial w_{11}^1}(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
&= a_1^1(1-a_1^1)x_1
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^1}{\partial w_{12}^1} &= \frac{\partial}{\partial w_{12}^1}\sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
&= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1)(1 - \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1))\frac{\partial}{\partial w_{12}^1}(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
&= a_1^1(1 - a_1^1)x_2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_2^1}{\partial w_{21}^1} &= \frac{\partial}{\partial w_{21}^1}\sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1)(1 - \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1))\frac{\partial}{\partial w_{21}^1}(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&= a_2^1(1 - a_2^1)x_1
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_2^1}{\partial w_{22}^1} &= \frac{\partial}{\partial w_{22}^1}\sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1)(1 - \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1))\frac{\partial a_2^1}{\partial w_{22}^1}\sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&= a_2^1(1 - a_2^1)x_2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^1}{\partial b_1^1} &= \frac{\partial}{\partial b_1^1}\sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
&= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1)(1 - \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1))\frac{\partial}{\partial b_1^1}(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
&= a_1^1(1 - a_1^1)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_2^1}{\partial b_2^1} &= \frac{\partial}{\partial b_2^1}\sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1)(1 - \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1))\frac{\partial}{\partial b_2^1}(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&= a_2^1(1 - a_2^1)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^2}{\partial w_{11}^2} &= \frac{\partial}{\partial w_{11}^2}\sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)(1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2))\frac{\partial}{\partial w_{11}^2}(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2(1 - a_1^2)a_1^1
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^2}{\partial w_{12}^2} &= \frac{\partial}{\partial w_{12}^2}\sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)(1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2))\frac{\partial}{\partial w_{12}^2}(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2(1 - a_1^2)a_2^1
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^2}{\partial b_1^2} &= \frac{\partial}{\partial b_1^2}\sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)(1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2))\frac{\partial}{\partial b_1^2}(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2(1 - a_1^2)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^2}{\partial w_{11}^1} &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)(1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2))\frac{\partial}{\partial w_{11}^1}(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)x_1 w_{11}^2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^2}{\partial w_{12}^1} &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)(1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2))\frac{\partial}{\partial w_{12}^1}(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)x_2 w_{11}^2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^2}{\partial w_{21}^1} &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)(1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2))\frac{\partial}{\partial w_{21}^1}(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)x_1 w_{12}^2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^2}{\partial w_{22}^1} &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)(1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2))\frac{\partial}{\partial w_{22}^1}(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)x_2 w_{12}^2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^2}{\partial b_1^1} &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)(1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2))\frac{\partial}{\partial b_1^1}(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)w_{11}^2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^2}{\partial b_2^1} &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)(1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2))\frac{\partial}{\partial b_2^1}(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)w_{12}^2
\end{aligned}
$$

$$C = (a_1^2 - z_1)^2$$

$$\begin{aligned}
\frac{\partial C}{\partial w_{11}^2} &= \frac{\partial}{\partial w_{11}^2}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)\frac{\partial}{\partial w_{11}^2}(a_1^2 - z_1) \\
&= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial w_{12}^2} &= \frac{\partial}{\partial w_{12}^2}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)\frac{\partial}{\partial w_{12}^2}(a_1^2 - z_1) \\
&= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial b_1^2} &= \frac{\partial}{\partial w_{12}^2}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)\frac{\partial}{\partial w_{12}^2}(a_1^2 - z_1) \\
&= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial w_{11}^1} &= \frac{\partial}{\partial w_{11}^1}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)\frac{\partial}{\partial w_{11}^1}(a_1^2 - z_1) \\
&= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)x_1 w_{11}^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial w_{12}^1} &= \frac{\partial}{\partial w_{12}^1}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)\frac{\partial}{\partial w_{12}^1}(a_1^2 - z_1) \\
&= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)x_2 w_{11}^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial w_{21}^1} &= \frac{\partial}{\partial w_{21}^1}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)\frac{\partial}{\partial w_{21}^1}(a_1^2 - z_1) \\
&= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)x_1 w_{12}^2
\end{aligned}$$

$$
\begin{aligned}
\frac{\partial C}{\partial w_{22}^1} &= \frac{\partial}{\partial w_{22}^1}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)\frac{\partial}{\partial w_{22}^1}(a_1^2 - z_1) \\
&= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)x_2 w_{12}^2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial b_1^1} &= \frac{\partial}{\partial b_1^1}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)\frac{\partial}{\partial b_1^1}(a_1^2 - z_1) \\
&= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)w_{11}^2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial b_2^1} &= \frac{\partial}{\partial b_2^1}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)\frac{\partial}{\partial b_2^1}(a_1^2 - z_1) \\
&= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)w_{12}^2
\end{aligned}
$$

3.3 Cost - Simplified

$$
\begin{aligned}
a_1^1 &= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
a_2^1 &= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
a_1^2 &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
C &= (a_1^2 - z_1)^2 \\
\frac{\partial C}{\partial w_{11}^2} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1 \\
\frac{\partial C}{\partial w_{12}^2} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1 \\
\frac{\partial C}{\partial b_1^2} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2) \\
\frac{\partial C}{\partial w_{11}^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)x_1 w_{11}^2 \\
\frac{\partial C}{\partial w_{12}^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)x_2 w_{11}^2 \\
\frac{\partial C}{\partial w_{21}^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)x_1 w_{12}^2 \\
\frac{\partial C}{\partial w_{22}^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)x_2 w_{12}^2 \\
\frac{\partial C}{\partial b_1^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)w_{11}^2 \\
\frac{\partial C}{\partial b_2^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)w_{12}^2
\end{aligned}
$$

3.4 Backpropagation

$$
\begin{aligned}
a_1^1 &= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
a_2^1 &= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
a_1^2 &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^2}{\partial a_1^1} &= \frac{\partial}{\partial a_1^1}\sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)(1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2))\frac{\partial}{\partial a_1^1}(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2(1 - a_1^2)w_{11}^2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial a_1^2}{\partial a_2^1} &= \frac{\partial}{\partial a_2^1}\sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)(1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2))\frac{\partial}{\partial a_2^1}(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2(1 - a_1^2)w_{12}^2
\end{aligned}
$$

$$
C = (a_1^2 - z_1)^2
$$

$$
\begin{aligned}
\frac{\partial C}{\partial a_1^2} &= \frac{\partial}{\partial a_1^2}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)\frac{\partial}{\partial a_1^2}(a_1^2 - z_1) \\
&= 2(a_1^2 - z_1)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial a_1^1} &= \frac{\partial}{\partial a_1^1}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)\frac{\partial}{\partial a_1^1}(a_1^2 - z_1) \\
&= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)w_{11}^2 \\
&= \frac{\partial C}{\partial a_1^2}a_1^2(1 - a_1^2)w_{11}^2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial a_2^1} &= \frac{\partial}{\partial a_1^1}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)\frac{\partial}{\partial a_1^1}(a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)w_{12}^2 \\
&= \frac{\partial C}{\partial a_1^2}a_1^2(1 - a_1^2)w_{12}^2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial w_{11}^2} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1 \\
&= \frac{\partial C}{\partial a_1^2}a_1^2(1 - a_1^2)a_1^1
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial w_{12}^2} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1 \\
&= \frac{\partial C}{\partial a_1^2}a_1^2(1 - a_1^2)a_2^1
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial b_1^2} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2) \\
&= \frac{\partial C}{\partial a_1^2} a_1^2(1 - a_1^2)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial w_{11}^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)x_1 w_{11}^2 \\
&= \frac{\partial C}{\partial a_1^1} a_1^1(1 - a_1^1)x_1
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial w_{12}^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)x_2 w_{11}^2 \\
&= \frac{\partial C}{\partial a_1^1} a_1^1(1 - a_1^1)x_2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial w_{21}^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)x_1 w_{12}^2 \\
&= \frac{\partial C}{\partial a_2^1} a_2^1(1 - a_2^1)x_1
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial w_{22}^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)x_2 w_{12}^2 \\
&= \frac{\partial C}{\partial a_2^1} a_2^1(1 - a_2^1)x_2
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial b_1^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)w_{11}^2 \\
&= \frac{\partial C}{\partial a_1^1} a_1^1(1 - a_1^1)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial C}{\partial b_2^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)w_{12}^2 \\
&= \frac{\partial C}{\partial a_2^1} a_2^1(1 - a_2^1)
\end{aligned}
$$

3.5 Backpropagation - Simplified

$$
\begin{aligned}
a_1^1 &= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
a_2^1 &= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
a_1^2 &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
C &= (a_1^2 - z_1)^2 \\
\frac{\partial C}{\partial a_1^2} &= 2(a_1^2 - z_1) \\
\frac{\partial C}{\partial a_1^1} &= \frac{\partial C}{\partial a_1^2} a_1^2(1 - a_1^2) w_{11}^2 \\
\frac{\partial C}{\partial a_2^1} &= \frac{\partial C}{\partial a_1^2} a_1^2(1 - a_1^2) w_{12}^2 \\
\frac{\partial C}{\partial w_{11}^2} &= \frac{\partial C}{\partial a_1^2} a_1^2(1 - a_1^2) a_1^1 \\
\frac{\partial C}{\partial w_{12}^2} &= \frac{\partial C}{\partial a_1^2} a_1^2(1 - a_1^2) a_2^1 \\
\frac{\partial C}{\partial b_1^2} &= \frac{\partial C}{\partial a_1^2} a_1^2(1 - a_1^2) \\
\frac{\partial C}{\partial w_{11}^1} &= \frac{\partial C}{\partial a_1^1} a_1^1(1 - a_1^1) x_1 \\
\frac{\partial C}{\partial w_{12}^1} &= \frac{\partial C}{\partial a_1^1} a_1^1(1 - a_1^1) x_2 \\
\frac{\partial C}{\partial w_{21}^1} &= \frac{\partial C}{\partial a_2^1} a_2^1(1 - a_2^1) x_1 \\
\frac{\partial C}{\partial w_{22}^1} &= \frac{\partial C}{\partial a_2^1} a_2^1(1 - a_2^1) x_2 \\
\frac{\partial C}{\partial b_1^1} &= \frac{\partial C}{\partial a_1^1} a_1^1(1 - a_1^1) \\
\frac{\partial C}{\partial b_2^1} &= \frac{\partial C}{\partial a_2^1} a_2^1(1 - a_2^1)
\end{aligned}
$$

3.6 Backpropagation - Generalised

$$
\begin{aligned}
\frac{\partial C}{\partial a_k^{(l-1)}} &= \sum_j \frac{\partial C}{\partial a_j^{(l)}} a_j^{(l)}(1 - a_j^{(l)}) w_{jk}^{(l)} \\
\frac{\partial C}{\partial w_{jk}^{(l)}} &= \frac{\partial C}{\partial a_j^{(l)}} a_j^{(l)}(1 - a_j^{(l)}) a_k^{(l-1)} \\
\frac{\partial C}{\partial b_j^{(l)}} &= \frac{\partial C}{\partial a_j^{(l)}} a_j^{(l)}(1 - a_j^{(l)})
\end{aligned}
$$

4 Backpropagation

4.1 Column

$$a^{(l)} = \begin{bmatrix} a_0^{(l)} \\ a_1^{(l)} \\ \vdots \\ a_n^{(l)} \end{bmatrix} \qquad b^{(l)} = \begin{bmatrix} b_0^{(l)} \\ b_1^{(l)} \\ \vdots \\ b_n^{(l)} \end{bmatrix} \qquad w^{(l)} = \begin{bmatrix} w_{11}^{(l)} & w_{12}^{(l)} & \cdots & w_{1n}^{(l)} \\ w_{21}^{(l)} & w_{22}^{(l)} & \cdots & w_{2n}^{(l)} \\ \vdots & \vdots & \ddots & \vdots \\ w_{m1}^{(l)} & w_{m2}^{(l)} & \cdots & w_{mn}^{(l)} \end{bmatrix}$$

$$\begin{aligned} a^{(0)} &= x \\ z^{(l+1)} &= a^{(l)} \cdot w^{(l)} + b^{(l)} \\ a^{(l)} &= \sigma(z^{(l)}) \end{aligned}$$

$$\begin{aligned} C &= (a^{(l)} - y)^2 \\ \frac{\partial C}{\partial a_j^{(L)}} &= 2(a_j^{(L)} - y_j) \\ \frac{\partial C}{\partial a_k^{(l-1)}} &= \sum_j \frac{\partial C}{\partial a_j^{(l)}} \sigma^{-1}(z_j^{(l)}) w_{jk}^{(l)} \\ \frac{\partial C}{\partial w_{jk}^{(l)}} &= \frac{\partial C}{\partial a_j^{(l)}} \sigma^{-1}(z_j^{(l)}) a_k^{(l-1)} \\ \frac{\partial C}{\partial b_j^{(l)}} &= \frac{\partial C}{\partial a_j^{(l)}} \sigma^{-1}(z_j^{(l)}) \end{aligned}$$

4.2 Row

$$a^{(l)} = \begin{bmatrix} a_0^{(l)} & a_1^{(l)} & \cdots & a_m^{(l)} \end{bmatrix} \qquad b^{(l)} = \begin{bmatrix} b_0^{(l)} & b_1^{(l)} & \cdots & b_m^{(l)} \end{bmatrix}$$

$$w^{(l)} = \begin{bmatrix} w_{11}^{(l)} & w_{12}^{(l)} & \cdots & w_{1n}^{(l)} \\ w_{21}^{(l)} & w_{22}^{(l)} & \cdots & w_{2n}^{(l)} \\ \vdots & \vdots & \ddots & \vdots \\ w_{m1}^{(l)} & w_{m2}^{(l)} & \cdots & w_{mn}^{(l)} \end{bmatrix}$$

$$\begin{aligned} a^{(0)} &= x \\ z^{(l+1)} &= a^{(l)} \cdot w^{(l)} + b^{(l)} \\ a^{(l)} &= \sigma(z^{(l)}) \end{aligned}$$

$$
\begin{aligned}
C &= (a^{(l)} - y)^2 \\
\frac{\partial C}{\partial a_j^{(L)}} &= 2(a_j^{(L)} - y_j) \\
\frac{\partial C}{\partial a_k^{(l-1)}} &= \sum_j \frac{\partial C}{\partial a_j^{(l)}} \sigma^{-1}(z_j^{(l)}) w_{kj}^{(l)} \\
\frac{\partial C}{\partial w_{kj}^{(l)}} &= \frac{\partial C}{\partial a_j^{(l)}} \sigma^{-1}(z_j^{(l)}) a_k^{(l-1)} \\
\frac{\partial C}{\partial b_j^{(l)}} &= \frac{\partial C}{\partial a_j^{(l)}} \sigma^{-1}(z_j^{(l)})
\end{aligned}
$$